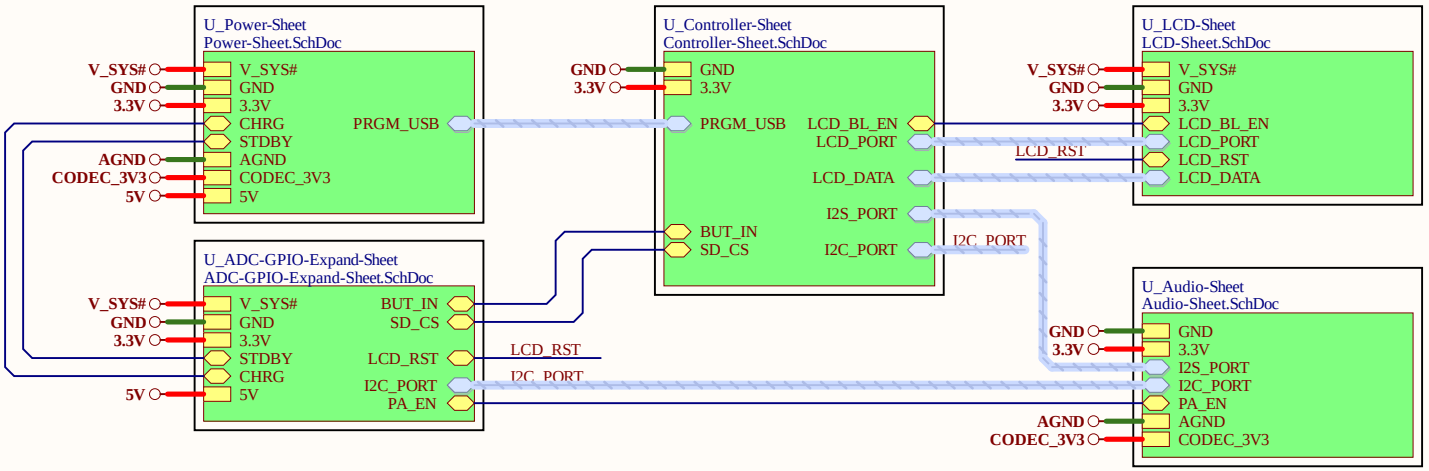
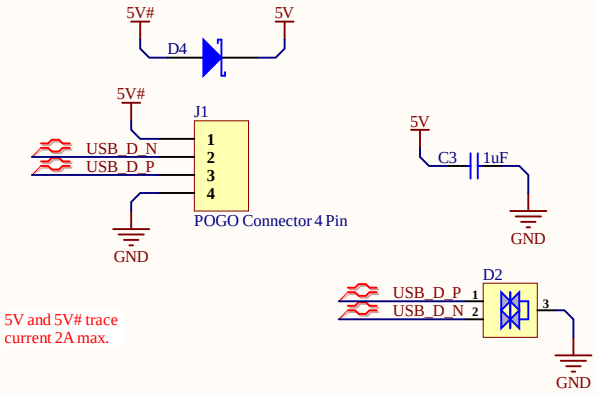


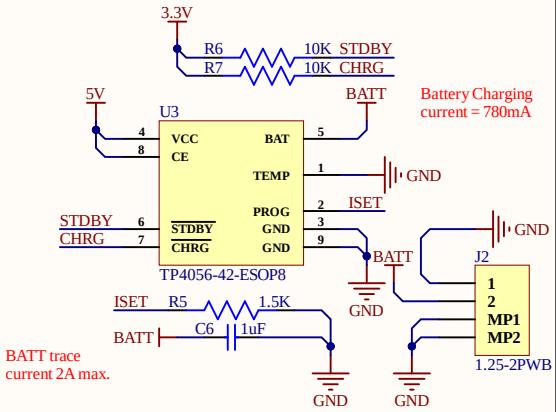
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Title: Top-Sheet.SchDoc	
File: Alarm-Clock-v1.0.PrjPcb	
Date: 2/15/2023	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 6

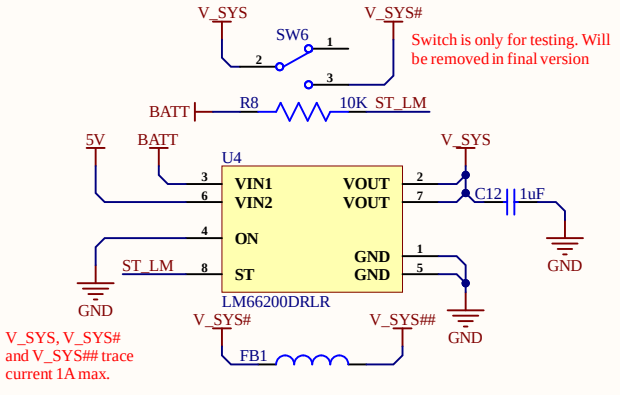
USB for Charging and Programming



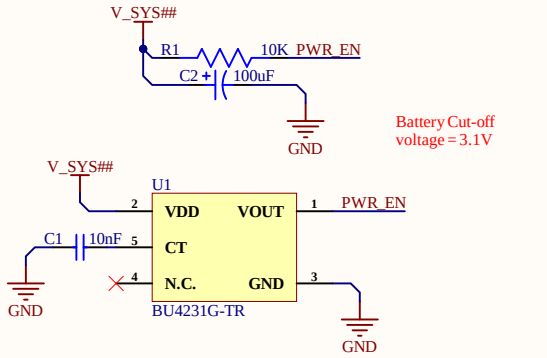
Battery Charger



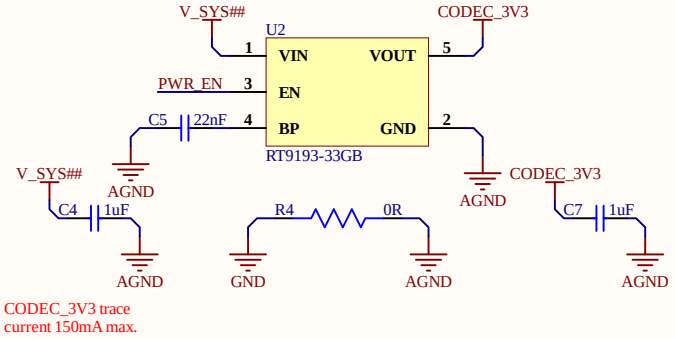
Power Priority Selection



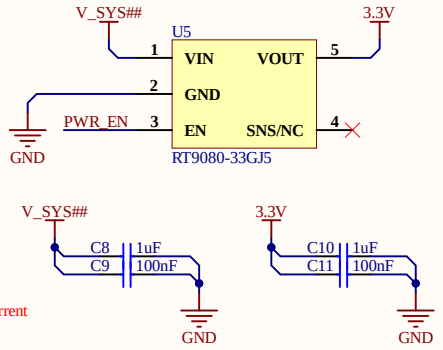
LOW Voltage Detector



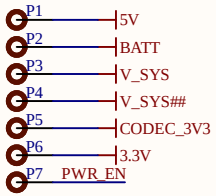
BAT to 3.3 Low Noise LDO

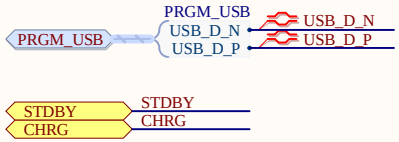


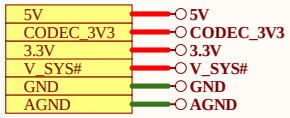
BAT to 3.3 LDO



Test Points

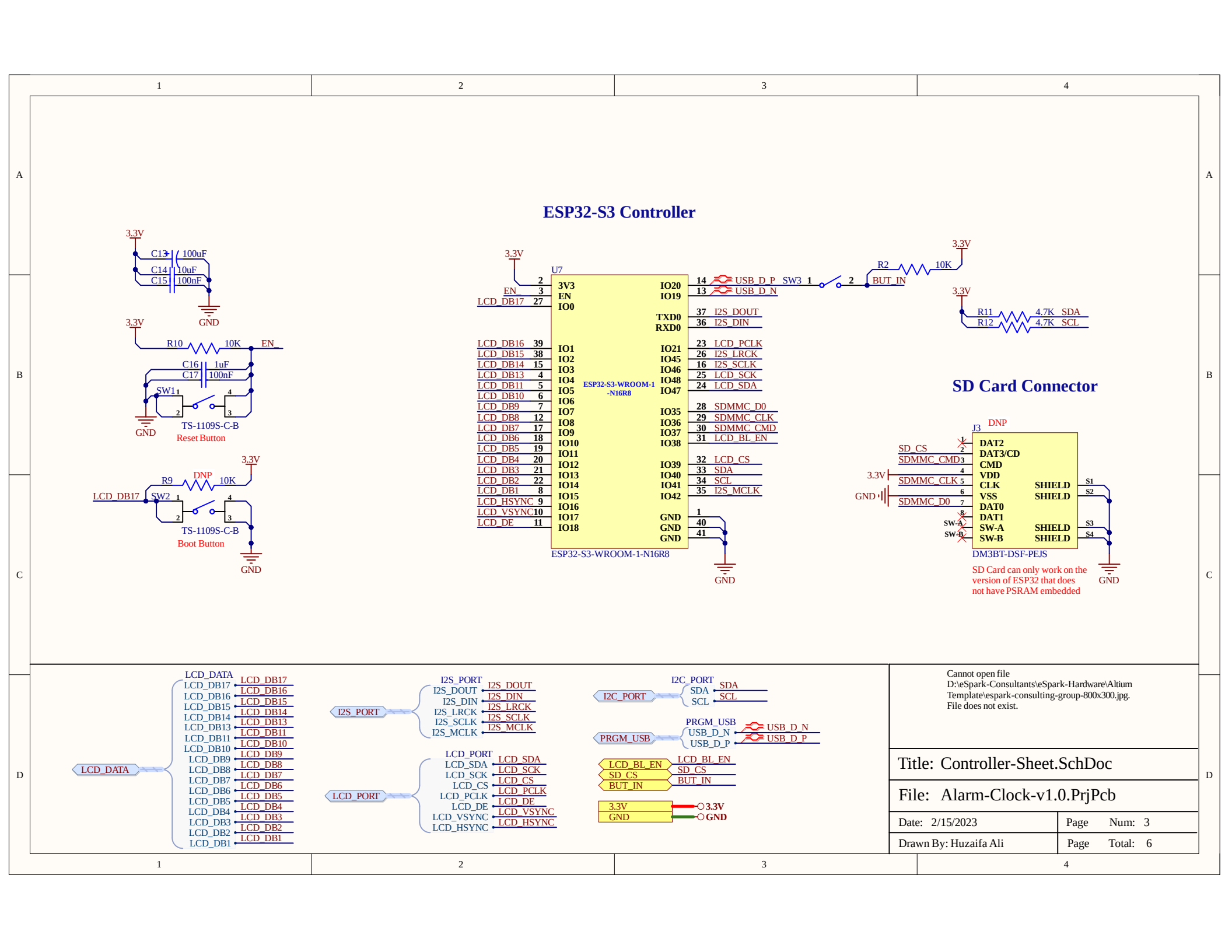






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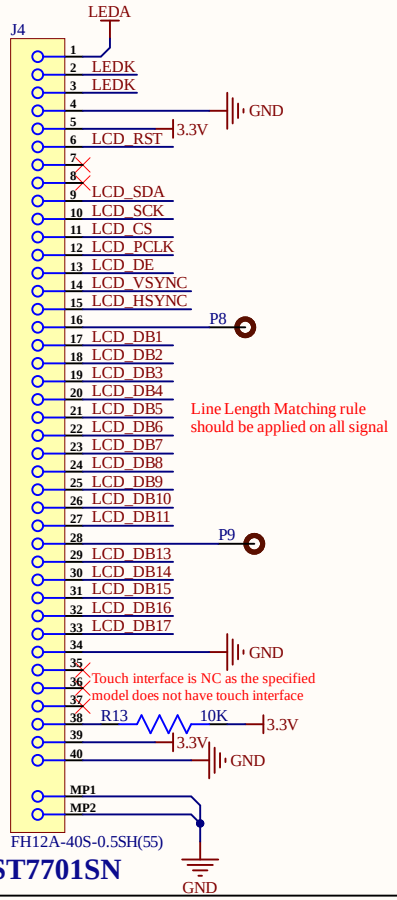
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Date: 2/15/2023	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 6



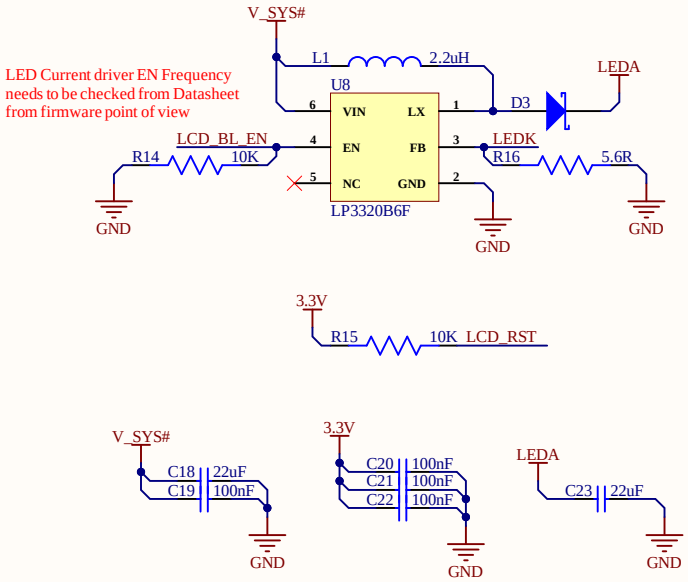
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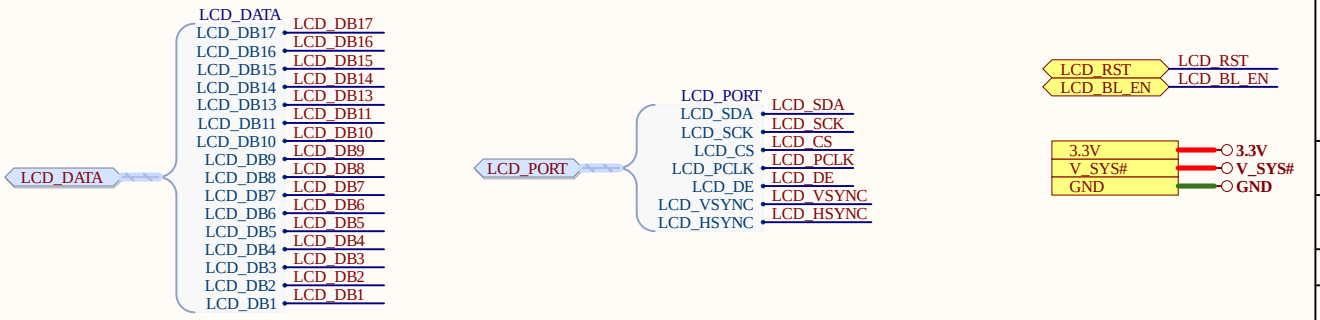
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File: Alarm-Clock-v1.0.PrjPcb	
Date: 2/15/2023	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 6

GX028-40RB-A1 LCD Connector



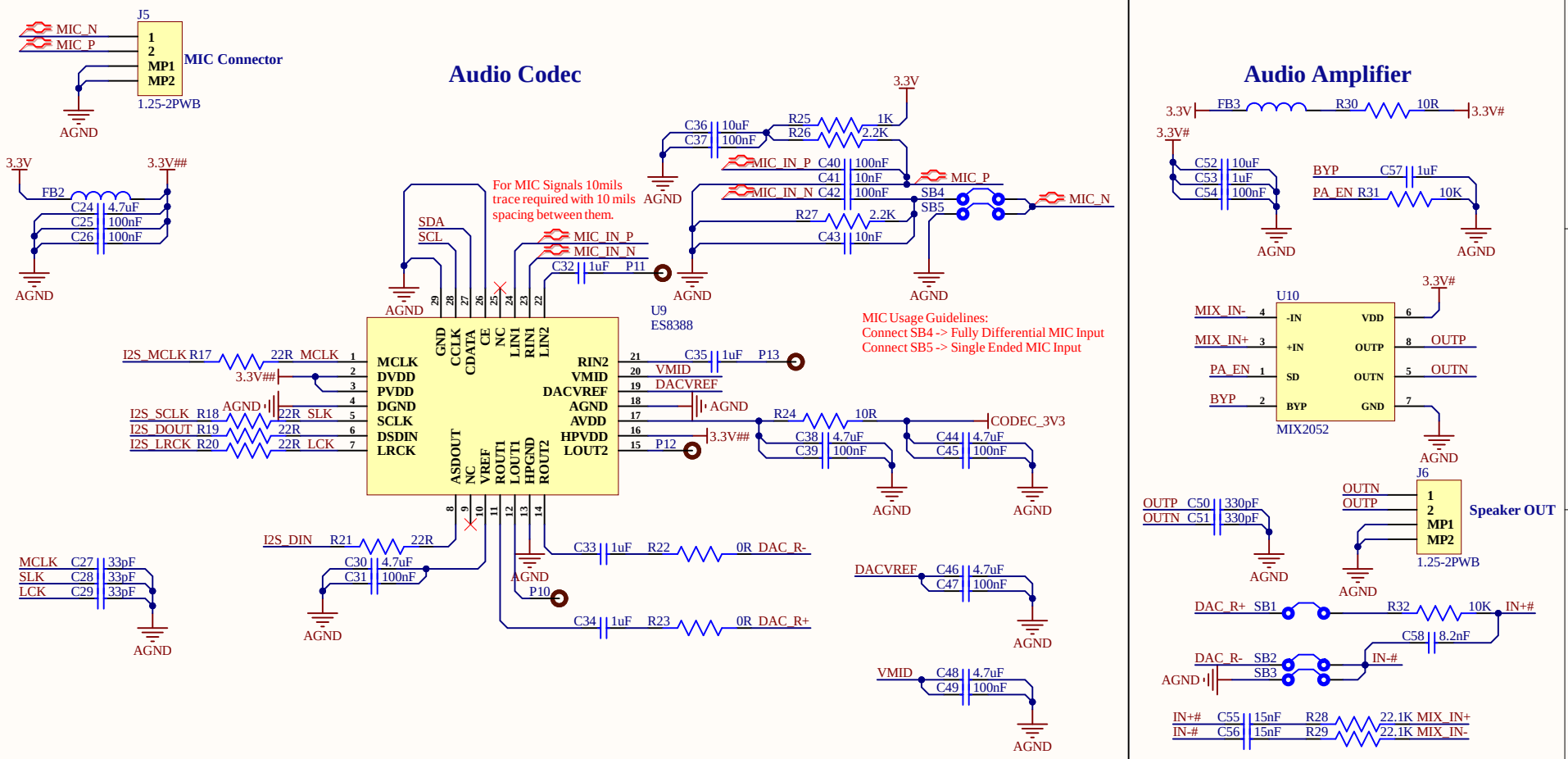
Constant Current LED Driver





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Template\espark-consulting-group-800x300.jpg.
File does not exist.

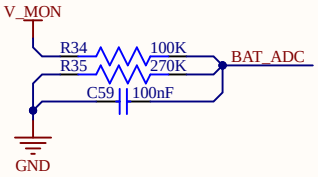
Title: LCD-Sheet.SchDoc	
File: Alarm-Clock-v1.0.PrjPcb	
Date: 2/15/2023	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 6


Audio Codec
J5
MIC_N
MIC_P
1
2
MP1
MP2
MIC Connector
1.25-2PWB
AGND
3.3V
3.3V##
FB2
C24 4.7uF
C25 100nF
C26 100nF
For MIC Signals 10mils trace required with 10 mils spacing between them.
SDA
SCL
MIC_IN_P
MIC_IN_N
C32 1uF P11
U9
ES8388
C36 10uF
C37 100nF
R25 1K
R26 2.2K
MIC_IN_P C40 100nF
C41 10nF
MIC_IN_N C42 100nF
R27 2.2K
C43 10nF
SB4
SB5
MIC_P
MIC_N
MIC Usage Guidelines:
Connect SB4 -> Fully Differential MIC Input
Connect SB5 -> Single Ended MIC Input
I2S_MCLK R17 22R MCLK
I2S_SCLK R18 22R SLK
I2S_DOUT R19 22R
I2S_LRCK R20 22R LCK
MCLK
DVDD
PVDD
DGND
SCLK
DSDIN
LRCK
RIN2
VMID
DACVREF
AGND
AVDD
HPVDD
LOUT2
C35 1uF P13
R24 10R
C38 4.7uF
C39 100nF
C44 4.7uF
C45 100nF
CODEC_3V3
P12
I2S_DIN R21 22R
C30 4.7uF
C31 100nF
C33 1uF R22 0R DAC_R-
C34 1uF R23 0R DAC_R+
P10
MCLK C27 33pF
SLK C28 33pF
LCK C29 33pF
DACVREF C46 4.7uF
C47 100nF
VMID C48 4.7uF
C49 100nF
Audio Amplifier
3.3V
FB3
R30 10R
3.3V#
C52 10uF
C53 1uF
C54 100nF
BYP C57 1uF
PA_EN R31 10K
U10
MIX_IN-
MIX_IN+
PA_EN
BYP
-IN
+IN
SD
BYP
VDD
OUTP
OUTN
GND
MIX2052
OUTP
OUTN
J6
1
2
MP1
MP2
Speaker OUT
OUTP C50 330pF
OUTN C51 330pF
DAC_R+ SB1
R32 10K IN+#
C58 8.2nF
DAC_R- SB2
SB3
IN-#
IN+# C55 15nF
IN-# C56 15nF
R28 22.1K MIX_IN+
R29 22.1K MIX_IN-
I2S_PORT
I2S_DOUT
I2S_DIN
I2S_LRCK
I2S_SCLK
I2S_MCLK
I2C_PORT
SDA
SCL
PA_EN
CODEC_3V3
3.3V
GND
AGND
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.
Title: Audio-Sheet.SchDoc
File: Alarm-Clock-v1.0.PrjPcb
Date: 2/15/2023
Page Num: 5
Drawn By: Huzaifa Ali
Page Total: 6

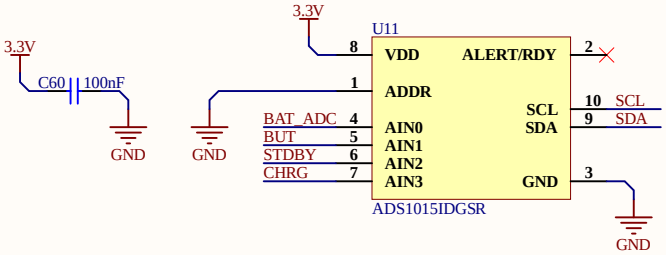
Battery voltage range
3.2V - 4.2V => 2.33V - 3.06V

Assuming battery capacity
1200mAH
4.2V = 100%
4.1V = 95%
4.0V = 84%
3.9V = 74%
3.8V = 60%
3.7V = 51%
3.6V = 37%
3.5V = 25%
3.4V = 18%
3.3V = 9%
3.2V = 0%

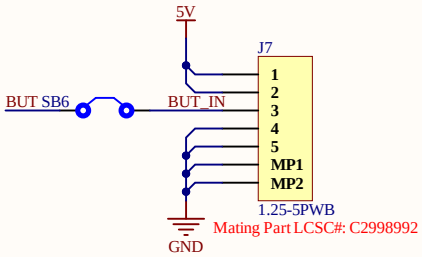
Detect Battery Voltage



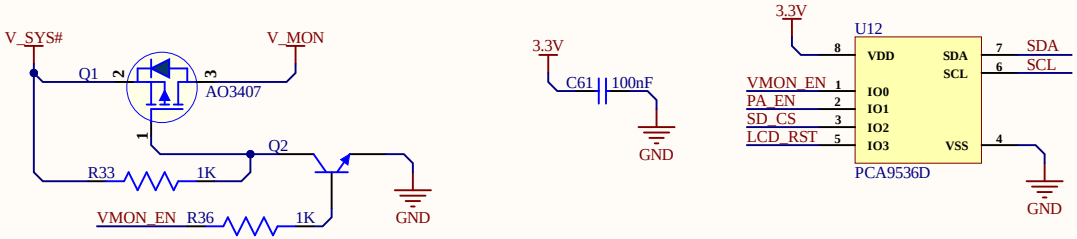
12-bit 4-Channel ADC

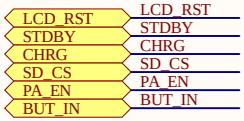


Side PCB Connector

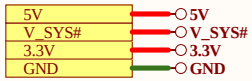


4-Channel GPIO Expander



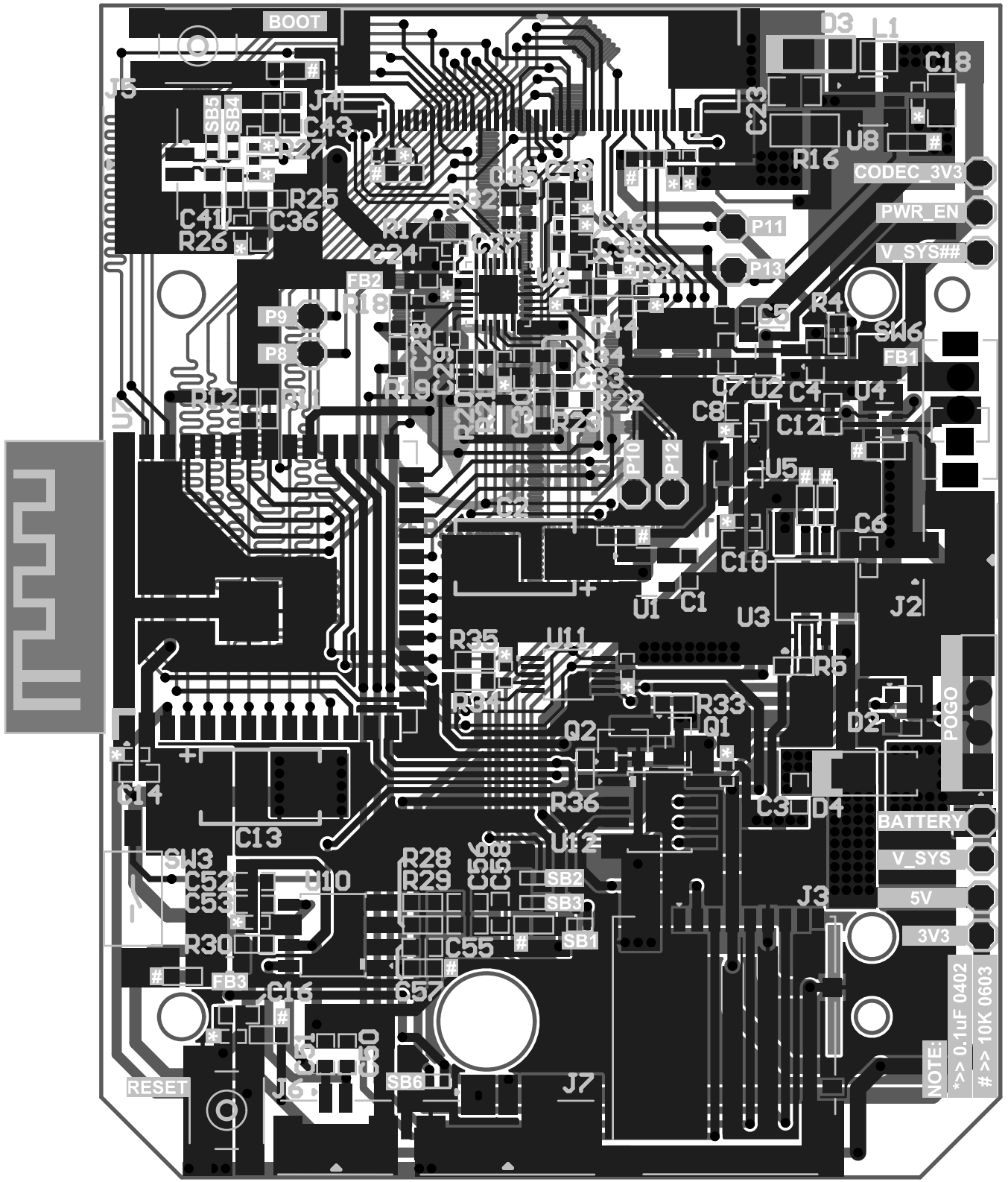






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Title: ADC-GPIO-Expand-Sheet.SchDoc	
File: Alarm-Clock-v1.0.PrjPcb	
Date: 2/15/2023	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 6


BOOT
D3
L1
C18
J5
J4
SB5
SB4
C43
R27
C23
U8
R16
CODEC_3V3
C35
C42
C16
C38
R25
C41
C36
R17
C24
R26
P11
PWR_EN
P13
V_SYS##
FB2
R18
P9
P8
C28
C29
C5
R4
SW6
FB1
C34
C33
R19
R20
R21
R22
R23
C30
C7
U2
C4
U4
C8
C12
P10
P12
U5
C6
C10
C1
U1
U3
J2
R35
U11
R5
R34
R33
Q1
Q2
D2
POGO
C14
R36
C3
D4
C13
U12
BATTERY
SW3
C52
C53
U10
R28
R29
C50
C56
SB2
SB3
SB1
V_SYS
5V
3V3
J3
R30
C55
FB3
C16
C57
RESET
J6
SB6
J7
NOTE:
*=> 0.1uF 0402
#=> 10K 0603

Comment	Description	Designator	Footprint	LibRef	Quantity
10nF	CAP CER 10000PF 50V C0G/NPO 0603	C1, C41, C43	Cap-0603	CC0603JRNPO9BN103	3
100uF	CAP TANT 100UF 20% 25V 2917	C2, C13	Case-E	TAJE107M025RNJ	2
1uF	CAP CER 1UF 25V X5R 0603	C3, C4, C6, C7, C8, C10, C12, C16, C32, C33, C34, C35, C53, C57	Cap-0603	CC0603KRX5R8BB105	14
22nF	CAP CER 0.022UF 50V X7R 0603	C5	Cap-0603	CC0603KRX7R9BB223	1
100nF	CAP CER 0.1UF 25V X5R 0402	C9, C11, C15, C17, C19, C20, C21, C22, C25, C26, C31, C37, C39, C40, C42, C45, C47, C49, C54, C59, C60, C61	Cap-0402	CC0402KRX5R8BB104	22
10uF	CAP CER 10UF 16V X5R 0603, CAP CER 10UF 10V X5R 0603	C14, C36, C52	Cap-0603	CC0603KRX5R7BB106, CC0603KRX5R6BB106	3
22uF	CAP CER 22UF 35V X5R 0805	C18, C23	Cap-0805	C2012X5R1V226MT000E	2
4.7uF	CAP CER 4.7UF 16V X5R 0603	C24, C30, C38, C44, C46, C48	Cap-0603	CC0603KRX5R7BB475	6
33pF	CAP CER 33PF 25V C0G/NPO 0402	C27, C28, C29	Cap-0402	SRM1555C1E330JA01D	3
330pF	CAP CER 330PF 50V C0G/NPO 0603	C50, C51	Cap-0603	CC0603JRNPO9BN331	2
15nF	CAP CER 0.015UF 50V X7R 0603	C55, C56	Cap-0603	CC0603JRX7R9BB153	2
8.2nF	CAP CER 8.2NF 50V X7R 10% 0603	C58	Cap-0603	CC0603KRX7R9BB822	1
ESD5302F-3/TR	16A 20V 7V 5V@Max Unidirectional SOT-23 TVS ROHS	D2	Sot-23	ESD5302F-3/TR	1
DSK24 K24	40V 550mV@2A 2A SOD-123FL Schottky Barrier Diodes (SBD) ROHS	D3	SOD-123FL	DSK24 K24	1
S532 /SMA(DO-214AC)	20V 500mV@3A 3A SMA(DO-214AC) Schottky Barrier Diodes (SBD) ROHS	D4	SMA	S532 /SMA(DO-214AC)	1
BLM18SG121TN1D	FERRITE BEAD 120 OHM 0603 1LN	FB1, FB2	IND-0603 (1608 Metric)	BLM18SG121TN1D	2
BLM18PG181SN1D	0603 180Ω ±25% DCR=0.08Ω 1.5A	FB3	IND-0603 (1608 Metric)	BLM18PG181SN1D	1
POGO Connector 4 Pin	Pogo Pad 2.5mm 4 Pin	J1	Pogo 4 Pin	POGO Connector 4 Pin	1
1.25-2PWB	1x2P 1 1.25mm Male pin 2 -40°C~+85°C Surface Mount SMD,P=1.25mm Wire To Board / Wire To Wire Connector ROHS	J2, J5, J6	1.25-2PWB	1.25-2PWB	3
DM3BT-DSF-PEJS	CONN MICRO SD CARD PUSH PUSH R/A	J3	DM3BT-DSF-PEJS	DM3BT-DSF-PEJS	1
FH12A-40S-0.5SH(55)	CONN FFC TOP 40POS 0.50MM R/A	J4	FH12A-40S-0.5SH(55)	FH12A-40S-0.5SH(55)	1
1.25-5PWB	1x5P 1 1.25mm Male pin 5 -25°C~+85°C Surface Mount SMD,P=1.25mm Wire To Board / Wire To Wire Connector ROHS	J7	1.25-5PWB	1.25-5PWB	1
2.2uH	FIXED IND 2.2UH 2A 158 MOHM SMD	L1	IHHP0806AZER2R2M01	IHHP0806AZER2R2M01	1
TEST POINT	Test Point ON	P1, P2, P3, P4, P5, P6, P7, P8, P9, P10, P11, P12, P13	Test-point	TEST POINT	13
AO3407	30V 4.1A 60mΩ@10V, 4.1A 350mW 3V@250uA P Channel SOT-23 MOSFETs ROHS	Q1	SOT-23	AO3407	1
MMBT2222A-7-F	TRANS NPN 40V 0.6A SMD SOT23-3	Q2	SOT-23	MMBT2222A-7-F	1
10K	RES SMD 10K Ohm 1% 1/10W 0603	R1, R2, R6, R7, R8, R9, R10, R13, R14, R15, R31, R32	Res-0603	RC0603FR-0710KL	12
0R	RES 0 OHM JUMPER 1/10W 0603	R4, R22, R23	Res-0603	RC0603JR-070RL	3
1.5K	RES 1.5K OHM 1% 1/10W 0603	R5	Res-0603	RC0603FR-071K5L	1
4.7K	RES SMD 4.7K OHM 5% 1/10W 0603	R11, R12	Res-0603	RC0603JR-074K7L	2
5.6R	General Purpose Chip Resistor Thick Film 1206 5.6Ohm 1% Paper T/R	R16	Res-1206	RC1206FR-7W5R6L	1
22R	RES 22 OHM 1% 1/10W 0603	R17, R18, R19, R20, R21	Res-0603	RC0603FR-0722RL	5
10R	RES SMD 10 OHM 1% 1/10W 0603	R24, R30	Res-0603	RC0603FR-0710RL	2
1K	RES SMD 1K OHM 1% 1/10W 0603	R25, R33, R36	Res-0603	RC0603FR-071KL	3
2.2K	RES 2.2K OHM 1% 1/10W 0603	R26, R27	Res-0603	RC0603FR-072K2L	2
22.1K	RES 22.1K OHM 1% 1/10W 0603	R28, R29	Res-0603	RC0603FR-0722K1L	2
100K	RES SMD 100K OHM 0.1% 1/10W 0603	R34	Res-0603	RT0603BRD07100KL	1
270K	RES 270K OHM 0.5% 1/10W 0603	R35	Res-0603	RC0603DR-07270KL	1
Solder Bridge 0402	Solder Bridge ON	SB1, SB2, SB3, SB4, SB5, SB6	Solder Bridge 0402	Solder Bridge 0402	6
TS-1109S-C-B	No NO 50mA 4.5mm 100000 260gf 12V -30°C~+85°C 4.5mm 4.3mm Black Round Button Brick nogging SPST SMD Tactile Switches ROHS	SW1, SW2	TS-1109S-C-B	TS-1109S-C-B	2
DSC01LSGET	SPST Black Flat dial, raised type 2.54mm 1 SMD,2.5x6mm DIP Switches ROHS	SW3	DSC01LSGET	DSC01LSGET	1
SS-12D02-G040	Straight 1A SPDT 8.7mm 8.6mm 24V 4.5mm Black PCPin 10000 Rectangular columnar -25°C~+70°C Plugin Slide Switches ROHS	SW6	SS-12D02-G040	SS-12D02-G040	1
BU4231G-TR	Power Supply Support Circuit, Fixed, 1 Channel, +3.1V, CMOS, PDSO5	U1	SSOP-5-1.6mm	BU4231G-TR	1
RT9193-33GB	IC REG LDO 3.3V 0.3A SOT23-5	U2	SOT-23-5	RT9193-33GB	1
TP4056-42-ESOP8	ESOP-8 Battery Management ICs ROHS	U3	ESOP-8	TP4056-42-ESOP8	1
LM66200DRLR	OR Controller N+1 ORing Controller N - Channel 2:1 SOT-583	U4	SOT-583	LM66200DRLR	1
RT9080-33GJ5	TSOT-23-5 Linear Voltage Regulators (LDO) ROHS	U5	TSOT-23-5	RT9080-33GJ5	1
ESP32-S3-WROOM-1-N16R8	SMD MODULE, ESP32-S3R8 WITH 3.3V	U7	41-SMD Module	ESP32-S3-WROOM-1-N16R8	1
LP3320B6F	SOT-23-6 AC-DC Controllers & Regulators ROHS	U8	SOT-23-6	LP3320B6F	1
ES8388	QFN-28-EP(4x4) ADC/DAC - Specialized ROHS	U9	QFN-28-EP(4x4)	ES8388	1
MIX2052	SOP-8 Audio Power OpAmps ROHS	U10	SOP-8	MIX2052	1
ADS1015IDGSR	IC ADC 12BIT SIGMA-DELTA 10VSSOP	U11	VSSOP-10	ADS1015IDGSR	1
PCA9536D	4.400kHz I²C, SMBus SOIC-8 I/O Expanders ROHS	U12	SOIC-8	PCA9536D	1